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00:00:00,000 --> 00:00:21,620
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00:00:21,620 --> 00:00:35,560
If we may get started, please.

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00:00:35,560 --> 00:00:47,440
It's my pleasure this afternoon to introduce Professor Roger Rovellet,
Professor of Science

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00:00:47,440 --> 00:00:52,880
and Public Policy, University of California, San Diego.

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00:00:52,880 --> 00:01:00,280
Roger is a former director of the Scripps Institute of Oceanography, 14
years, science

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00:01:00,280 --> 00:01:04,920
advisor to the Secretary of the Interior.

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00:01:04,920 --> 00:01:12,040
He's a member of the National Academy of Sciences, and he's contributed
in the international

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00:01:12,040 --> 00:01:17,640
area, for example, being a part of the U.S. delegation to the United
Nations Conference

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00:01:17,640 --> 00:01:21,720
on Science and Technology for Development.

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00:01:21,720 --> 00:01:28,240
His recent articles appear in journals such as Energy and Climate,
Science, Scientific

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00:01:28,240 --> 00:01:29,240
American.

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00:01:29,240 --> 00:01:38,280
They've treated topics, for example, global population and aspects of
food problems, the

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00:01:38,280 --> 00:01:48,200

carbon cycle and the biosphere, energy use in rural India, ocean drilling, and assistance

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00:01:48,200 --> 00:01:52,440

a climatologist can lend to developing countries.

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00:01:52,440 --> 00:01:59,320

We're looking forward, and I'm looking forward, to his talk on the role of the oceans in Earth's

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00:01:59,320 --> 00:02:03,200

climate and carbon budget.

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00:02:03,200 --> 00:02:07,920

Professor Rovellet.

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00:02:07,920 --> 00:02:26,640

I'm going to talk mostly today about the carbon dioxide problem, which is a subject of considerable

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00:02:26,640 --> 00:02:35,000

interest to at least one component of this laboratory.

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00:02:35,000 --> 00:02:39,680

It's possible to say, I think, that from the standpoint of living creatures on this planet,

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00:02:39,680 --> 00:02:47,200

and particularly, of course, from the standpoint of human beings, that the most important substance

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00:02:47,200 --> 00:02:53,400

there is, the most important chemical compound there is, is carbon dioxide.

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00:02:53,400 --> 00:02:55,520

That's true for three reasons.

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00:02:55,520 --> 00:03:01,760

In the first place, we're all made up of carbon and a few other things, and all of

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00:03:01,760 --> 00:03:10,360

that carbon comes to us by way of plants, which in turn extract carbon from the carbon

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00:03:10,360 --> 00:03:22,200

dioxide in the air and combine it with water to form sugars and other related substances.

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00:03:22,200 --> 00:03:28,280

Life as we know it, or as we can even imagine it, would be impossible without carbon dioxide.

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00:03:28,280 --> 00:03:39,680

Second, from the standpoint of us animals, as opposed to plants, an essential substance

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00:03:39,680 --> 00:03:49,200

for us is free oxygen, oxygen to facilitate our very high level of metabolism.

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00:03:50,200 --> 00:03:56,600

If you ever stop to think about a tree and why a tree is so sedentary, it's because a

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00:03:56,600 --> 00:04:00,880

tree has so little energy.

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00:04:00,880 --> 00:04:07,000

A tree that tried to walk around would quickly collapse from lack of energy.

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00:04:07,000 --> 00:04:16,880

We're among the most energy-using of all animals, in fact, one kilogram of human flesh radiates

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00:04:16,880 --> 00:04:25,120

about 10,000 times as much energy per second as an equal kilogram of the sun.

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00:04:25,120 --> 00:04:33,800

And this very high level of energy use depends upon the oxidation of organic matter in our

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00:04:33,800 --> 00:04:40,360

metabolic systems, and that oxygen, which is essential for our high metabolism, probably

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00:04:40,360 --> 00:04:46,840

all come from the reduction of carbon by plant photosynthesis, reduction of carbon

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00:04:46,840 --> 00:04:53,000

dioxide by plant photosynthesis, and the freeing of oxygen.

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00:04:53,000 --> 00:05:00,040

Third, another essential requirement for living creatures is the presence of liquid water.

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00:05:00,040 --> 00:05:06,440

As far as we know, we're the only planet in our solar system that possesses liquid water

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00:05:06,440 --> 00:05:08,720

on its surface.

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00:05:08,720 --> 00:05:11,800

And part of the reason for this, of course, is that we're just about at the right distance

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00:05:11,800 --> 00:05:12,800

from the sun.

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00:05:12,800 --> 00:05:19,760

But an equally important reason is that the presence of carbon dioxide in the atmosphere

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00:05:19,760 --> 00:05:26,200

produces something which has been called the greenhouse effect.

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00:05:26,200 --> 00:05:32,880

Because the carbon dioxide absorbs infrared radiation, it scatters it back to the lower

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00:05:32,880 --> 00:05:33,880

atmosphere.

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00:05:33,880 --> 00:05:39,560

The temperature near the Earth's surface is about 30 degrees centigrade higher than it

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00:05:39,560 --> 00:05:42,160

would be in the absence of carbon dioxide.

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00:05:42,160 --> 00:05:48,200

A 30-degree drop in the surface temperature of the Earth would mean that the entire Earth

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00:05:48,200 --> 00:05:54,160

was covered with ice and that there wouldn't be any liquid water on it.

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00:05:54,160 --> 00:06:00,560

So the carbon dioxide plays three essential roles, more than essential, fundamental roles

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00:06:00,640 --> 00:06:02,640

in our very existence.

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00:06:02,640 --> 00:06:09,640

At the same time, we're now talking about something called the carbon dioxide problem.

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00:06:09,640 --> 00:06:16,640

And what I want to do today is to talk about this carbon dioxide problem.

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00:06:17,360 --> 00:06:22,440

We could define it, not perhaps in a word or a sentence, but in a paragraph.

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00:06:22,440 --> 00:06:28,160

Very accurate, continuous measurements of the carbon dioxide content in the air at two

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00:06:28,160 --> 00:06:35,160

isolated stations, in Mauna Loa in the Hawaiian Islands, at the top of Mauna Loa in the Hawaiian

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00:06:35,560 --> 00:06:41,840
Islands, and at the South Pole.

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00:06:41,840 --> 00:06:47,840
These measurements have been made since about 1957, something over 20 years.

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00:06:47,840 --> 00:06:54,160
And they show beyond any doubt that the carbon dioxide content of the air has increased by

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00:06:54,160 --> 00:07:01,160
about 60 percent during the past 20 years, and that the rate of increase is accelerating.

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00:07:05,920 --> 00:07:11,400
It is equally clear that human activities, the burning of coal, oil, and natural gas,

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00:07:11,400 --> 00:07:16,040
and the clearing of forests for agricultural and other purposes, are the major sources

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00:07:16,040 --> 00:07:19,600
of the added carbon dioxide.

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00:07:19,600 --> 00:07:25,200
If the worldwide economy, with its insatiable demand for energy, continues to grow and fossil

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00:07:25,200 --> 00:07:31,040
fuels continue to be the principal source of energy, the carbon dioxide content of the

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00:07:31,040 --> 00:07:38,040
air can be expected to double before the year 2000, before the year 2050, excuse me.

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00:07:39,720 --> 00:07:45,600
That is, within the lifetimes of many human beings who are now alive.

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00:07:45,600 --> 00:07:51,680

Because carbon dioxide in the atmosphere absorbs and re-radiates infrared radiation,

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00:07:51,680 --> 00:07:57,000

the temperature of the lower atmosphere will rise perhaps to a greater extent than at any

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00:07:57,000 --> 00:08:03,360

time since the end of the last ice age, about 10,000 years ago.

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00:08:03,360 --> 00:08:08,740

Profound changes in the world climate might occur, with correspondingly great impacts

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00:08:08,740 --> 00:08:15,740

on human societies, and the effects are likely to persist for many centuries thereafter,

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00:08:17,060 --> 00:08:24,060

once we've embarked on this path of increasing carbon dioxide and its possible accompanying

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00:08:25,100 --> 00:08:31,580

climatic changes. These climatic changes will persist, certainly for 500 years, and perhaps

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00:08:31,580 --> 00:08:33,700

for a longer time.

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00:08:33,700 --> 00:08:39,780

This is the carbon dioxide problem. This problem is virtually unique among today's human-induced

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00:08:39,780 --> 00:08:46,780

problems. Heaven knows we have enough of them. But this is a unique problem in three ways,

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00:08:47,260 --> 00:08:54,260

or not entirely unique, somewhat unique. First, its long timescale. Second, its uncertainty.

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00:08:55,260 --> 00:09:02,260

And third, its planetary character. Like the movement of a glacier, which is barely detectable

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00:09:04,620 --> 00:09:10,460

from year to year, but has very great cumulative effects over many years, the effects of the

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00:09:10,460 --> 00:09:16,540

annual increase in atmospheric carbon dioxide are imperceptibly small. We're none of us

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00:09:16,540 --> 00:09:23,540

aware of it in any physiological way. And in fact, the effects in terms of climate are

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00:09:24,380 --> 00:09:30,300

change, potential climate change, have so far been imperceptible, compared with the

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00:09:30,300 --> 00:09:36,100

noise level. The signal of the carbon dioxide signal is below the noise level of inter-annual

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00:09:36,100 --> 00:09:43,100

fluctuations in climate. But the effects are likely to be accumulating from year to year,

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00:09:44,340 --> 00:09:51,180

and in the long run, they're likely to prove both large and inexorable. Human societies

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00:09:51,180 --> 00:09:58,180

have no memory of a comparable climatic event in their recorded histories. Although the

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00:09:58,380 --> 00:10:04,560

coming and going of the last ice age was an event of even larger magnitude, its memory

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00:10:04,560 --> 00:10:10,180

is submerged far below consciousness among today's human beings. We may still have some

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00:10:10,180 --> 00:10:17,180
kind of a racial memory of the last ice age, but it doesn't manifest
itself at the surface

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00:10:17,580 --> 00:10:21,260
of consciousness.

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00:10:21,260 --> 00:10:26,820
Uncertainty is the second fundamental characteristic of the carbon
dioxide problem. And before

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00:10:26,820 --> 00:10:33,820
I talk about these uncertainties, let me first say a word about the
certainties. I mentioned,

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00:10:35,060 --> 00:10:39,020
to begin with, the measurements that have been made on the top of Mauna
Loa and at the

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00:10:39,020 --> 00:10:46,020
South Pole. And I'll try to draw these on the board.

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00:10:47,580 --> 00:10:54,580
If this is CO₂ in the air, and this is time, and let's suppose we start
at 1957, when the

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00:11:00,220 --> 00:11:07,220
measurements began, and go up to 1977, which is the last year for which
we have well-analyzed

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00:11:09,060 --> 00:11:16,060
data, the measurements were made by a group of scientists, and the
measurements were made

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00:11:17,300 --> 00:11:24,300
by a rather fancy infrared analyzer that took the infrareds, measured the
infrared absorption

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00:11:25,260 --> 00:11:32,020
in continuous air samples. So the measurements were made by a
continuously recording device.

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00:11:32,020 --> 00:11:39,020

And what they show was that you have a carbon dioxide content in the air at the top of Mauna

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00:11:39,020 --> 00:11:47,020

Loa that looks like this. And if we draw a curve through this, it shows that the carbon

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00:12:10,020 --> 00:12:11,820

dioxide content is at the top of Mauna Loa. And if I draw this correctly, it would be

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00:12:11,820 --> 00:12:18,820

an exponential curve. Slightly curving upward. These fluctuations, which are on both sides

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00:12:31,380 --> 00:12:38,140

of the average of the curve, are annual fluctuations, and they depend upon the fact that in the

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00:12:38,140 --> 00:12:45,140

summertime in the northern hemisphere, there's a good deal of photosynthesis, which withdraws

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00:12:46,420 --> 00:12:52,400

carbon dioxide from the air. In the wintertime, in the northern hemisphere, there's a good

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00:12:52,400 --> 00:12:58,780

deal more respiration than there is photosynthesis, and carbon dioxide is released to the air.

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00:12:58,780 --> 00:13:05,780

So these seasonal fluctuations in the carbon dioxide are about three parts per million.

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00:13:06,300 --> 00:13:13,300

This is, let's say, 300 parts per million, and up here is 330. Well, let's say this is

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00:13:17,540 --> 00:13:24,540

a little bit more than 300. This is about 310, and this is about 330. These annual fluctuations

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00:13:25,260 --> 00:13:32,260

are about three parts per million, and that's, as I said, due to the seasonal differences

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00:13:32,980 --> 00:13:39,980

between photosynthesis and respiration. We find exactly the same kind of curve at the

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00:13:39,980 --> 00:13:46,980

South Pole, but with the annual variations being less, and a lag of about two years in

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00:13:48,260 --> 00:13:55,260

the total amount in the air, compared to the measurements at Mauna Loa. These are both

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00:13:55,780 --> 00:14:02,780

quite understandable. First place in the southern hemisphere, there's much less photosynthesis

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00:14:02,820 --> 00:14:08,340

because the southern hemisphere is mostly ocean, and also correspondingly much less

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00:14:08,340 --> 00:14:15,340

respiration. But more particularly, the lag is due to the fact that the added carbon dioxide

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00:14:19,540 --> 00:14:24,540

is coming primarily from human activities in the northern hemisphere, where the lands

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00:14:24,540 --> 00:14:31,540

are concentrated. And there's a window, sort of a wall or barrier at the equator across

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00:14:32,260 --> 00:14:39,260

which the atmosphere doesn't exchange very rapidly or very readily. And so it takes about

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00:14:39,900 --> 00:14:46,900

two years for the South Pole Station to catch up with the Mauna Loa Station. The accuracy

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00:14:47,900 --> 00:14:54,900

of these measurements is about a tenth of a part per million. And if we look at this

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00:14:56,460 --> 00:15:03,460

curve here in some detail, it shows wiggles. I didn't draw the wiggles, but it's not an

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00:15:08,860 --> 00:15:15,860

absolutely straight line. Sometimes there's a somewhat higher increase in the atmosphere

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00:15:16,620 --> 00:15:23,620

and sometimes a somewhat lower increase, but overall a continuous increase. That's the

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00:15:25,220 --> 00:15:32,220

only certainty in this morass of uncertainties that I want to talk about today. However,

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00:15:35,420 --> 00:15:41,580

quite a while ago, the great Swedish chemist Svante Arrhenius and the American geologist

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00:15:41,580 --> 00:15:48,580

Thomas C. Chamberlain in the 1890s postulated that the burning of coal by the growing rate

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00:15:53,820 --> 00:16:00,820

at which coal was being burnt by the expanding and developing industrial civilization would

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00:16:03,100 --> 00:16:10,100

very likely increase the CO₂ content of the air, and that this in turn could have an effect

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00:16:10,140 --> 00:16:17,140

on climate in raising the temperature of the lower air. A man named Callender, about 1940,

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00:16:19,500 --> 00:16:26,500

said that the observed increase in global average temperatures over the Earth, particularly

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00:16:26,900 --> 00:16:33,060

in the Northern Hemisphere, which had been going on from 1870, for about 70 years the

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00:16:33,060 --> 00:16:39,060

global temperature in the Northern Hemisphere increased slowly but more or less steadily.

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00:16:39,660 --> 00:16:46,660

Callender said this could very well be due to the added carbon dioxide in the air. The

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00:16:49,500 --> 00:16:56,500

measurements made, however, in the 1890s and even the measurements made in the 1930s and

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00:16:57,540 --> 00:17:04,540

early 1940s were not very accurate. They had probably the 19th century measurements might

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00:17:05,340 --> 00:17:12,340

be in error by as much as 10 or 20 parts per million. The measurements even in the 1930s,

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00:17:12,620 --> 00:17:19,620

early 1940s, probably were in error by a few parts per million. It wasn't until the development

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00:17:21,260 --> 00:17:28,260

of the infrared analyzer in the late 1940s and early 1950s that very accurate measurements

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00:17:29,260 --> 00:17:36,260

could be made. And these are the measurements that have been made at Mauna Loa and at the

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00:17:36,980 --> 00:17:43,980

South Pole by a great chemical analyst, Charles David Keeling. You said I was going to talk

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00:17:44,700 --> 00:17:50,700

about the ocean, Dr. Batesill. I will talk about the Scripps Institution of Oceanography

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00:17:50,700 --> 00:17:57,420

in the sense that Dr. Keeling was brought to Scripps in 1957 to do that, to make these

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00:17:57,420 --> 00:18:01,980

measurements as part of the International Geophysical Year. And he's been making them

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00:18:01,980 --> 00:18:08,740

ever since and he's still at Scripps. So that's at least one way in which the ocean affects

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00:18:08,740 --> 00:18:15,460

the carbon dioxide problem. He wants to measure carbon dioxide in his belly. He never really

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00:18:15,460 --> 00:18:20,860

wanted to do anything else. He measures it with extreme accuracy and extreme precision

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00:18:20,860 --> 00:18:25,620

and thinks about all the things that might possibly go wrong with all the measurements

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00:18:25,620 --> 00:18:32,620

related to carbon dioxide. Really a remarkable individual. And he's produced this remarkable

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00:18:35,980 --> 00:18:42,980

result, this analytical result. And a great pyramid of speculation and uncertainty has

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00:18:45,580 --> 00:18:52,580

been built upon this result. For example, we know that since, say, 1850, about 130 years

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00:18:53,580 --> 00:19:00,580

ago, there were probably around 130 billion tons of carbon released by the burning of

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00:19:03,100 --> 00:19:10,100

coal, oil, and natural gas. And that, of course, resulted in the production of carbon dioxide

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00:19:11,100 --> 00:19:18,100

and water. Nothing particularly notable about the water, but the carbon dioxide produced

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00:19:22,620 --> 00:19:28,420

by this combustion of coal, oil, and natural gas was, until fairly recently, thought to

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00:19:28,420 --> 00:19:35,420

be the principal source of the added CO₂ in the air. And if we compute on the basis of

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00:19:35,740 --> 00:19:42,740

Keeling's measurements between 1957 and 1977, what proportion of the added carbon dioxide

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00:19:47,740 --> 00:19:54,740

produced by fossil fuel combustion stayed in the air and was found at Mount Loa? It's

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00:19:54,820 --> 00:20:01,820

of the order of perhaps 50 percent, about half of the, or perhaps a little bit more

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00:20:02,820 --> 00:20:09,820

than half of the added carbon dioxide has stayed in the air. Here we come against one

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00:20:11,820 --> 00:20:17,220

of the first of our uncertainties. We don't really know what the carbon dioxide content

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00:20:17,220 --> 00:20:23,260

was in the middle of the 19th century. So we don't know, before 1957, what fraction

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00:20:23,260 --> 00:20:30,260

of the CO₂ produced by the burning of fossil fuels remained in the air. There's a man named

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00:20:30,500 --> 00:20:37,500

Binzie Stoiver at the University of Washington who, by measurements of carbon-13, carbon-12

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00:20:37,700 --> 00:20:44,700

isotope ratios, has concluded that, in fact, the CO₂ content at that time was considerably

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00:20:44,740 --> 00:20:51,740

lower than it was usually assumed to be. In other words, more CO₂ has been added than

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00:20:52,740 --> 00:20:59,740

you would compute on the basis of Keeling's measurements. This has raised the problem

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00:21:02,780 --> 00:21:09,780

of where does the additional CO₂ come from. Most biologists, at least most American biologists,

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00:21:13,860 --> 00:21:19,340

are convinced that forests are being cut down at a rate greater than they are growing in

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00:21:19,340 --> 00:21:26,340

the world as a whole. I, myself, am convinced that this has been true ever since 1850.

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00:21:30,380 --> 00:21:37,380

The combustion of wood, the combustion of the wood stored in the forest, and the reduction

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00:21:38,620 --> 00:21:43,420

of the organic matter in the forest soils, or the oxidation of the organic matter in

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00:21:43,420 --> 00:21:50,420

the forest soils, that these have both contributed perhaps as much as a half as much carbon dioxide

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00:21:50,740 --> 00:21:57,740

as the fossil fuel combustion has contributed. We then are faced with an even more serious

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00:21:58,460 --> 00:22:04,820

problem. Where has the carbon dioxide content gone? Where has the added carbon gone, carbon

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00:22:04,820 --> 00:22:11,820

dioxide gone? There are two schools of thought about this. One of them is that it's going

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00:22:13,540 --> 00:22:20,540

right back into the biosphere. Carbon dioxide added to the air should result in an increase

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00:22:23,140 --> 00:22:29,060

in photosynthesis. It acts just like a fertilizer, just as nitrogen or phosphate does, and it's

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00:22:29,060 --> 00:22:36,060

just as essential for plant growth as water. So it could very well be that photosynthesis

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00:22:37,060 --> 00:22:44,060

has increased in the uncut forest, which means that the trees are getting bigger. That would

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00:22:44,820 --> 00:22:50,460

seem like a very simple thing to find out about, if the trees are getting bigger. The

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00:22:50,460 --> 00:22:54,100

trouble is that where the trees are getting bigger is probably mostly in the tropical

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00:22:54,100 --> 00:23:01,100

rainforest. The trees in the tropical rainforest have no tree rings, because there's no difference

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00:23:01,940 --> 00:23:07,020

between summer and winter. The result is that it's essentially impossible to tell whether

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00:23:07,020 --> 00:23:14,020

the trees are bigger or not. The alternative explanation, or the alternative hypothesis

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00:23:15,460 --> 00:23:21,700

as to where the carbon dioxide has gone is that the oceanographers are wrong about the

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00:23:21,700 --> 00:23:25,700

amount that's gone into the ocean, and that much more has gone into the ocean than they

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00:23:25,700 --> 00:23:32,700

think has been the case. Naturally, I take a dim view of this second hypothesis, and

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00:23:34,940 --> 00:23:41,940

for fairly good reasons, I think. I think that the biosphere has not been either much

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00:23:43,500 --> 00:23:50,500

of a source or much of a sink for carbon dioxide. But these questions about where the carbon

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00:23:51,060 --> 00:23:58,060

dioxide has come from and where it's gone, how much of that produced has remained in

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00:23:58,660 --> 00:24:05,660

the air, are of course a first-order uncertainty, because if we're going to predict how much

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00:24:06,860 --> 00:24:13,860

future CO₂ will be in the air, we have to have some notion of how much stays there out

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00:24:14,460 --> 00:24:19,260

of the total amount produced. There is, however, an even more fundamental uncertainty, and

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00:24:19,260 --> 00:24:26,260

that is how much will be produced. During the past 125 years, the rate of combustion

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00:24:26,380 --> 00:24:33,380

of fossil fuels has gone up about 4 percent a year, exponentially, with a doubling time

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00:24:33,700 --> 00:24:40,700

of around 20 years. As long as that doubling time of 20 years persists, everybody will

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00:24:41,340 --> 00:24:48,340

agree that more and more carbon dioxide will remain in the air. There's a larger and larger

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00:24:53,420 --> 00:25:00,420

proportion or fraction of that produced that will remain in the air. But there's no reason

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00:25:01,140 --> 00:25:07,060

whatever to believe that we're going to continue using fossil fuels at 4 percent a year, with

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00:25:07,060 --> 00:25:14,060

an increasing use rate of 4 percent a year, an exponentially increasing use rate. It's

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00:25:14,860 --> 00:25:21,860

just as likely that in fact we may have reached the peak of fossil fuel use. That is, because

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00:25:23,540 --> 00:25:30,540

of the disappearance of cheap oil, the appearance of very expensive energy, that the energy

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00:25:31,540 --> 00:25:38,540

uses may decline markedly. And that would be particularly true if we have a worldwide

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00:25:39,700 --> 00:25:46,700

depression and a large amount of unemployment and human misery. The carbon dioxide problem

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00:25:49,100 --> 00:25:56,100

would then be substituted for by a ghastly, hair-curling economic depression. It's possible

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00:25:57,100 --> 00:26:04,100

that there will be some intermediate between no increasing use of fossil fuels and the

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00:26:04,300 --> 00:26:10,660

4 percent increase. The lower the rate of use of fossil fuels, the longer the time will

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00:26:10,660 --> 00:26:17,660

be before we double or quadruple the carbon dioxide content of the air. The doubling or

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00:26:18,660 --> 00:26:25,660

quadrupling, nevertheless, would be inexorable and inevitable if use rates continue at say

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00:26:31,540 --> 00:26:38,540

even 2 percent. Growth of fossil fuels continues at even 2 percent a year. The third major

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00:26:39,540 --> 00:26:46,540

kind of uncertainty is what will be the effect of the additional CO₂ in the air? Supposing

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00:26:48,260 --> 00:26:55,260

we double carbon dioxide content, what will be the effect on the climate? In my opinion,

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00:26:57,900 --> 00:27:04,900

the best estimates of these effects have been made by Siyakuro Manabe and his colleagues

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00:27:05,580 --> 00:27:12,580

at the Geophysical Fluid Dynamics Laboratory in Princeton, New Jersey, a NOAA laboratory.

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00:27:13,980 --> 00:27:18,980

Manabe thinks that with a doubling of the carbon dioxide content, that the global average

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00:27:18,980 --> 00:27:25,980

temperature will rise about 2 degrees Centigrade. The temperature in the northern tier of states

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00:27:27,580 --> 00:27:34,580

in the United States will rise about 5 degrees Centigrade. The average CO2 in the northern

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00:27:34,900 --> 00:27:41,900

tier will rise about 7 or 8 degrees Centigrade. These are changes in temperature, as I said

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00:27:54,900 --> 00:28:01,900

to begin with, larger than any we've experienced. Excursions in average global or regional temperature

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00:28:02,380 --> 00:28:09,380

are larger than any that have been experienced for the last 10,000 years. Perhaps even more

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00:28:09,740 --> 00:28:16,740

serious would be that these latitudinal differences in the temperature increase would result also

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00:28:20,660 --> 00:28:27,660

in a profound change in the regional patterns of precipitation. Some regions would become

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00:28:28,660 --> 00:28:34,660

much drier, some would become much wetter. Which regions would become drier and which

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00:28:34,660 --> 00:28:41,660

would become wetter, I believe is still almost a completely unknown problem. The climatic

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00:28:46,660 --> 00:28:52,660

effects increase more or less as the logarithm of the CO2 content of the air, so a quadrupling

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00:28:52,660 --> 00:28:59,460

of CO2 would not result in another doubling of temperature or another 2 degrees increase

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00:28:59,460 --> 00:29:06,340

in temperature, but a somewhat smaller increase than that. Nevertheless, this quadrupling

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00:29:06,340 --> 00:29:13,340

is very likely to occur by the end of the next century, with even larger climatic effects

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00:29:14,340 --> 00:29:21,340

than those that I have specified. It's only fair to say that there's very considerable

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00:29:23,260 --> 00:29:30,260

uncertainty about these climatic models. The uncertainties arise from two principal causes.

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00:29:30,440 --> 00:29:35,660

One of them is that it's possible that the cloudiness of the Earth would increase and

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00:29:35,660 --> 00:29:41,180

that the clouds would increase in such a way that the albedo, that is the reflectivity

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00:29:41,180 --> 00:29:48,180

of the Earth's surface, would increase. If we have low clouds over a larger area of the

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00:29:49,660 --> 00:29:56,220

Earth than we have now, then that certainly would be very likely, that would be almost

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00:29:56,220 --> 00:30:02,060

surely the outcome. The reflectivity of the Earth would increase, less solar radiation

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00:30:02,060 --> 00:30:06,860

would be absorbed in the surface and the ocean and the ground, more would be reflected to

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00:30:06,860 --> 00:30:12,140

outer space. And this would certainly be a negative feedback, which would reduce the

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00:30:12,140 --> 00:30:18,140
climate effect. If, on the other hand, the clouds are high clouds, just the reverse would

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00:30:18,140 --> 00:30:22,860
occur, because the high clouds have such a relatively low temperature that the radiation

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00:30:22,860 --> 00:30:29,500
from them is lower, the infrared radiation from them is lower than that near the Earth's

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00:30:29,500 --> 00:30:34,980
surface, and that in effect would tend to warm the lower atmosphere and add to the carbon

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00:30:34,980 --> 00:30:40,580
dioxide effect. So we don't really know whether the clouds will be good or bad, whether there

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00:30:40,580 --> 00:30:45,940
will be a negative or a positive feedback. The one thing that does seem certain is that

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00:30:45,940 --> 00:30:52,940
there will be more evaporation, therefore the relative humidity of the atmosphere will

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00:30:53,540 --> 00:30:58,900
remain constant. If the relative humidity of the atmosphere remains constant, the absolute

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00:30:59,220 --> 00:31:05,500
humidity will increase, which means that precipitation overall, the global precipitation, will also

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00:31:05,500 --> 00:31:12,500
increase. We don't know what will happen to the sea ice in the northern latitudes, and

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00:31:14,020 --> 00:31:18,700
particularly Manabe's calculation indicates that the Arctic ice would disappear in the

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00:31:18,700 --> 00:31:25,700

summertime, the sea ice covering the Arctic Ocean, with undoubtedly profound but not at

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00:31:26,700 --> 00:31:33,460

all understood effects on the global climate, basically because of the very great change

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00:31:33,460 --> 00:31:40,460

in the regime of heat exchange between the Arctic Ocean and the atmosphere, if there

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00:31:40,460 --> 00:31:47,460

is no blanket of ice insulating the ocean from the atmosphere. The other great uncertainty

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00:31:48,460 --> 00:31:55,460

in the models is that the models at the present time don't take the ocean into account in

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00:31:59,860 --> 00:32:06,860

any realistic way. The atmospheric modelers have invented two kinds of oceans. In one

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00:32:09,100 --> 00:32:14,940

kind they treat the ocean as if it were a swamp, which means that it has no heat capacity

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00:32:14,980 --> 00:32:21,980

but is a source of water to the atmosphere. Water exchange is not at all impeded. The

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00:32:22,500 --> 00:32:29,500

other kind of ocean is an ocean covered with a copper plate, in other words an ocean with

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00:32:30,500 --> 00:32:36,060

an infinite heat capacity, so that the temperature of the ocean cannot change no matter what

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00:32:36,060 --> 00:32:43,060

happens to the atmosphere. Neither the copper plate ocean nor the swamp ocean give it all

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00:32:43,780 --> 00:32:50,780

a realistic picture of the nature of the great atmospheric ocean climate machine.

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00:32:53,660 --> 00:33:00,660

Just to give you a case in point, about a year ago we heard rather widely spread rumors

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00:33:02,540 --> 00:33:09,540

that the British Meteorological Office had made calculations, had a big climatic model

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00:33:10,300 --> 00:33:14,580

and had made calculations that showed that the carbon dioxide effect would be negligibly

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00:33:14,580 --> 00:33:21,580

small. But it turned out when we got to the World Climate Conference in Geneva and John

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00:33:22,580 --> 00:33:29,260

Mason, the director of that office, who is a typical Englishman, always tends to play

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00:33:29,260 --> 00:33:36,260

down any natural phenomenon that might disturb the British Raj, the British climate model

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00:33:39,740 --> 00:33:45,100

had to confess that their model, which they used to demonstrate that it would be no effect,

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00:33:45,100 --> 00:33:49,980

in fact they kept the temperature of the ocean constant, which means of course there could

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00:33:49,980 --> 00:33:56,460

not be any effect. When they had varied the temperature of the ocean, when they had raised

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00:33:56,460 --> 00:34:03,460

it two degrees, then they got the same results that Manabe got. So this was a punctured balloon

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00:34:03,900 --> 00:34:10,900

as far as the climate modelers are concerned. The other uncertainty, apart from the uncertainty

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00:34:13,740 --> 00:34:20,740

in the models, is the uncertainty as to what will be the pattern of precipitation. There

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00:34:22,180 --> 00:34:29,180

is some evidence from the so-called climatic optimum, which occurred right at the end of

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00:34:30,060 --> 00:34:35,420

the ice age, when the temperature was perhaps one or two degrees warmer than it is now,

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00:34:35,420 --> 00:34:42,420

that North America would have a considerable decrease in precipitation and that other areas,

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00:34:44,060 --> 00:34:50,500

for example the Sahelian area of the Sahara, would have a considerable increase, just based

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00:34:50,500 --> 00:34:57,500

upon where the precipitation seems to have occurred at the time of the climatic optimum.

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00:34:58,020 --> 00:35:03,460

But that climatic optimum may or may not have been a warming of the atmosphere due to added

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00:35:03,460 --> 00:35:08,740

carbon dioxide in the air. It might have been due to some other cause. So we can't really

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00:35:08,740 --> 00:35:15,740

take those fragmentary and uncertain results too seriously.

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00:35:18,980 --> 00:35:25,980

The third or fourth area of uncertainty is what will be the effects. I want to talk about

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00:35:26,980 --> 00:35:33,060

some of the effects and the uncertainties regarding them. Primarily what will be the

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00:35:33,060 --> 00:35:40,060

effects on things about our environment which are of direct and one might almost say vital

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00:35:41,500 --> 00:35:48,500

interest to human beings. Let me start with the ocean. Here I would like to combine both

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00:35:48,500 --> 00:35:55,500

the ocean and the ice, the ocean hydrosphere and the cryosphere. One of the obvious consequences

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00:36:06,380 --> 00:36:12,140

of the changes in temperature that I have described is that there would be less temperature

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00:36:12,140 --> 00:36:18,180

difference between low latitudes and high latitudes. That is, the high latitudes would

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00:36:18,180 --> 00:36:25,180

be relatively much warmer compared to their present average temperatures than the low

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00:36:26,780 --> 00:36:32,780

latitudes. So there would be a smaller temperature gradient, presumably therefore much less heat

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00:36:32,780 --> 00:36:38,660

transport between low latitudes and high latitudes, therefore much less wind. Because what drives

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00:36:39,660 --> 00:36:46,660

the atmospheric and oceanic circulations is the transport of heat from low latitudes to

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00:36:46,820 --> 00:36:53,700

high latitudes. Less heat transport, presumably less circulation. If the winds in the atmosphere

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00:36:53,700 --> 00:37:00,380

are lower, then the wind stress on the sea surface would be smaller. In fact, the wind

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00:37:00,380 --> 00:37:07,380

stress varies as the square of the wind strength. Therefore we presumably would have less heat

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00:37:08,660 --> 00:37:15,660

transfer, less stirring in the ocean, weaker currents, therefore less vertical as well

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00:37:19,220 --> 00:37:26,220

as horizontal motions in the sea. Specifically, probably a considerably smaller upwelling.

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00:37:27,900 --> 00:37:34,260

An upwelling of various kinds is the principal source of the world's fisheries. This might

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00:37:34,300 --> 00:37:41,300

have a devastating effect, or at least a serious effect, on the world's fisheries. I've already

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00:37:41,660 --> 00:37:45,900

mentioned the possibility that the Arctic ice might disappear. It might essentially

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00:37:45,900 --> 00:37:52,260

melt in the summertime. A prospect that frightens the glaciologists is that the West Antarctic

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00:37:52,260 --> 00:37:58,540

ice cap would melt, or not would melt, but would disappear. We had a conference on this

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00:37:58,540 --> 00:38:05,540

subject just last week in Maine, where people came from all over the world. I'm happy to

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00:38:06,220 --> 00:38:10,780

report to you that there are just about as many glaciologists who think that the ice

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00:38:10,780 --> 00:38:17,780

cap won't disappear as there are who think it will. So you can pay your money and take

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00:38:19,900 --> 00:38:26,300

your choice. However, it's possible that we can find out more about this and what their

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00:38:26,340 --> 00:38:32,420

principal task, in fact, was to invent a research program that would tell us more about what

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00:38:32,420 --> 00:38:39,420

the Antarctic ice cap might do. The problem is basically this, that the West Antarctic

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00:38:40,340 --> 00:38:47,340

ice cap is resting on seawater, not on solid ground. Parts of its bottom are way below

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00:38:47,940 --> 00:38:54,940

sea level. It seems to be held in place by the ice shelves that surround it, which essentially

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00:39:04,060 --> 00:39:11,060

hold it back. The ice shelves, in turn, are kept from melting by sea ice, by the Antarctic

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00:39:11,300 --> 00:39:16,620

sea ice, which extends out a considerable distance beyond the shelf. If the temperature

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00:39:16,620 --> 00:39:22,500

in the Antarctic latitudes increases by 5 to 10 degrees, one might suppose that the

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00:39:22,500 --> 00:39:29,500

sea ice would melt, the ice shelf would disappear, and the ice cap, the West Antarctic ice cap,

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00:39:31,540 --> 00:39:38,540

would grind into the ocean. Nothing would be holding it back. If this happened, the

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00:39:39,540 --> 00:39:46,540

sea level would rise about 6 meters, about 20 feet. This might happen in 100 years. The

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00:39:49,540 --> 00:39:55,940

reason why the alarmists think of this time constant is that apparently exactly this kind

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00:39:55,940 --> 00:40:02,940

of thing happened in Hudson's Bay at the end of the last ice age. The ice cap covering

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00:40:02,980 --> 00:40:09,980

Hudson's Bay seems to have disappeared in a matter of a few decades. I have no opinion

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00:40:11,220 --> 00:40:18,220

about this. I would point out, however, that even this project, although it's clearly uncomfortable,

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00:40:20,100 --> 00:40:27,100

not quite as serious as it looks. Cities are rebuilt about once every 50 years. If you

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00:40:27,700 --> 00:40:30,940

look at the number of old buildings in New York, there are perhaps one out of 100 that

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00:40:30,940 --> 00:40:37,940

are more than 50 years old, or maybe one out of 50. What we do simply is build New

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00:40:42,220 --> 00:40:49,220

York somewhere else. Similarly with Miami, and with London, and possibly with Oakland.

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00:40:51,180 --> 00:40:58,180

It may not be such a bad thing.

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00:40:58,620 --> 00:41:05,620

If we look at the effect on the land, we can think of the effects here in different directions.

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00:41:10,140 --> 00:41:16,580

First the effect on what we have called the unmanaged biosphere, that is, the grasslands,

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00:41:16,580 --> 00:41:23,580

the forests, the tundra, the savannas, the rainforests and the tropics, the boreal forests,

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00:41:23,580 --> 00:41:30,580

and so forth. This is a very complicated problem. Let me now return to what I said

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00:41:34,380 --> 00:41:41,380

earlier, and that is that adding carbon dioxide to the atmosphere will increase photosynthesis.

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00:41:43,580 --> 00:41:50,580

CO2 acts as a fertilizer. It may very well be that in some places, and it's not always

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00:41:53,620 --> 00:41:59,900

but sometimes, and with some plants, that carbon dioxide in the air is the limiting

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00:41:59,900 --> 00:42:06,900

factor for their growth. If that is the case, then an increase in carbon dioxide will cause

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00:42:07,100 --> 00:42:14,100

those plants to photosynthesize more rapidly. If you just think about this all by itself,

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00:42:15,220 --> 00:42:19,980

you might think that this would be on the whole a net benefit to the plants and to everybody

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00:42:20,020 --> 00:42:24,740

and all other forms of life that depend on the plants. Unfortunately, the problem is

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00:42:24,740 --> 00:42:31,300

not quite so simple. It's not quite so simple because of four kinds of problems. The first

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00:42:31,300 --> 00:42:38,300

problem is that an increase in atmospheric temperature accompanying the rise of carbon

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00:42:39,380 --> 00:42:45,540

dioxide would cause an increase in respiration, and it's by no means clear whether oxidation

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00:42:45,540 --> 00:42:51,620

of organic matter, the reverse of photosynthesis, would not proceed at a more rapid rate than

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00:42:51,620 --> 00:42:56,620

the increase in photosynthesis. In other words, we might have a negative effect because of

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00:42:56,620 --> 00:43:03,620

the increase in respiration. The second problem is that an increase in temperature would of

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00:43:07,780 --> 00:43:14,220

course cause a disequilibrium in the plants. They would tend to move north, into more northern

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00:43:14,420 --> 00:43:21,420

latitudes. Say, the spruce would tend to migrate 200 or 300 miles farther north, the center

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00:43:23,620 --> 00:43:30,020

of the spruce population, or the pine population, or the population of any particular species

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00:43:30,020 --> 00:43:37,020

of trees. If this happens, then something called phenology comes into effect. The plants

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00:43:38,020 --> 00:43:45,020

are very sensitive to the length of the day. They're often said to be photosensitive, and

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00:43:47,540 --> 00:43:54,540

plants flower, they drop their leaves, they do everything, and time it by day length.

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00:43:55,780 --> 00:44:00,540

If they move to a different latitude, the day length will be different. It'll be longer

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00:44:00,540 --> 00:44:06,540

in the summer and shorter in the winter than the day length that they're used to, and that

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00:44:06,540 --> 00:44:11,660

is very likely to seriously disturb their behavior. Some plants will be disturbed more

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00:44:11,660 --> 00:44:18,660

than others. Those plants that are least disturbed will prosper. Those plants that are most disturbed

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00:44:19,180 --> 00:44:26,180

will fall by the wayside. You change the character of the forest. You change the ecologic makeup

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00:44:26,300 --> 00:44:31,580

of the grasslands and of the forest and of the other areas.

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00:44:31,580 --> 00:44:38,580

Fourth, we tend to think of plants as sort of amorphous blobs of organic matter. We think

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00:44:39,860 --> 00:44:46,860

of a tree as capable of growing much bigger than it is, or a bush being bigger or littler

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00:44:48,060 --> 00:44:54,220

depending upon its environment. The biologists don't think this is so. They think that in

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00:44:54,220 --> 00:45:01,220

fact plants have a proper size and a proper shape, just like we do. They think that plants

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00:45:01,580 --> 00:45:08,580

do. If you got twice as much food, you might get fatter, but you wouldn't get any taller,

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00:45:08,860 --> 00:45:14,220

and you might not even get any fatter. We all have a proper size and shape. Insofar

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00:45:14,220 --> 00:45:21,220

as plants have also a proper size and shape, then they will not respond in any,Äthey may

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00:45:22,020 --> 00:45:27,260

respond in quite different ways than one can predict to a change in atmospheric carbon

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00:45:27,340 --> 00:45:32,900

dioxide. It's the combination of the change in atmospheric carbon dioxide and the change

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00:45:32,900 --> 00:45:39,900

in temperature, and therefore in the latitude for different plants, that fraughts this question

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00:45:42,900 --> 00:45:49,780

of the reaction of the natural biosphere, or freights it, whatever the present tense

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00:45:49,780 --> 00:45:56,780

of fraught is, that freights the question of the carbon dioxide in the natural biosphere

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00:45:58,260 --> 00:46:05,260

with great uncertainty. As far as the managed biosphere is concerned, there we have a quite

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00:46:05,780 --> 00:46:11,540

different situation. The United States agriculturalists are quite convinced that they can cope with

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00:46:11,540 --> 00:46:18,540

anything, particularly the research, the state experiments, the research workers at

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00:46:18,660 --> 00:46:24,300

the state experiment stations, and they point out with considerable justification that they

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00:46:24,420 --> 00:46:30,740

have in fact coped with a lot over the past 50 years or so as they've learned more about

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00:46:30,740 --> 00:46:37,740

the genetics and physiology of crop plants. Whether they can cope with a problem of the

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00:46:39,140 --> 00:46:45,860

following kind is not quite obvious. At the present time, the greatest food-producing

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00:46:45,860 --> 00:46:52,860

area in the world is the Corn Belt of Illinois, Iowa, Missouri, and Indiana. Here, the God-given

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00:46:54,340 --> 00:47:00,660

crop seems to have been, as he has in so many ways, extraordinarily kind to the United States.

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00:47:00,660 --> 00:47:07,100

The corn, the climate there, the temperature, the rainfall, the soil, the soil is rich

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00:47:07,100 --> 00:47:13,180

and deep, the rainfall is abundant, and the temperature is warm but not too warm, in such

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00:47:13,180 --> 00:47:20,180

a way that the yields of corn have continually increased until they're now about something

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00:47:21,180 --> 00:47:28,180

like 3 or 4 tons, well, about 4 tons per acre per year, average yields. Very, very

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00:47:31,180 --> 00:47:37,180

high yields, indeed. About 10 tons per hectare of corn, about 30 tons per hectare of organic

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00:47:37,180 --> 00:47:44,180

matter. Now, supposing we have a marked increase in temperature and a marked decrease in rainfall

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00:47:44,580 --> 00:47:49,860

in the Corn Belt, it might be necessary to move the Corn Belt into Canada, or at least

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00:47:49,860 --> 00:47:56,860

further north in the United States, into Michigan and Minnesota, northern Wisconsin.

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00:47:57,020 --> 00:48:03,260

Problem here is that the soils there are poor. Their podzols, their forest soils, have nowhere

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00:48:03,260 --> 00:48:09,060

near the fertility or the adaptability that the soils of Iowa and Illinois and Missouri

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00:48:09,060 --> 00:48:16,060

and Indiana have. So this might result in a very considerable decrease in production

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00:48:16,660 --> 00:48:22,780

of corn production just by having to move the Corn Belt. It may be impossible to genetically

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00:48:22,780 --> 00:48:29,780

modify the corn plants so they will still produce the very high yields in the areas

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00:48:30,180 --> 00:48:37,180

where they now grow. If we think about agriculture as far as the rest of the world is concerned,

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00:48:37,820 --> 00:48:41,140

it's probably true that all the developed countries will be in the same condition as

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00:48:41,140 --> 00:48:48,140

the United States. That is, that they will be able to cope or adapt to a climatic change

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00:48:49,740 --> 00:48:56,740

and take advantage of the change of the additional carbon dioxide. For example, if the experiments

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00:49:01,620 --> 00:49:08,620

in greenhouses are correct and photosynthesis does increase very considerably with added

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00:49:09,620 --> 00:49:16,620

carbon dioxide, and if genetic modifications can be made so that the plants will not respire

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00:49:16,620 --> 00:49:21,620

any more rapidly with higher temperatures, then you should get more crops. But we're

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00:49:21,620 --> 00:49:27,340

not really interested in larger crops. What we're interested in is more grain, more wheat

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00:49:27,340 --> 00:49:33,980

or corn or potatoes, whatever the edible part of the crops. What must be done and what I'm

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00:49:33,980 --> 00:49:38,620

sure can be done and what the agricultural people in the United States are convinced

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00:49:38,620 --> 00:49:45,620

can be done is to tinker with the genetics of the crop plants so in fact the added photosynthesis

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00:49:47,020 --> 00:49:54,020

and production of photosynthetic product or material gets into the seed and you increase

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00:49:54,060 --> 00:50:01,060
the seed content of the plants, which is what we eat and what we want. If we look at the

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00:50:01,340 --> 00:50:05,420
rest of the world, however, the underdeveloped countries where 90% of all the people are

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00:50:05,420 --> 00:50:12,420
going to be living by the end of this century, the position is very much more grim. It will

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00:50:14,100 --> 00:50:20,460
be more grim for those areas where the rainfall becomes less. It may become better for those

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00:50:20,460 --> 00:50:27,460
areas which become wetter, have a more uniform and constant rainfall. Please take note of

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00:50:31,060 --> 00:50:35,300
this word constant. One of the things that we don't know at all and which the climate

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00:50:35,300 --> 00:50:42,300
models don't tell us is what will be the variability in climate from year to year. If that variability

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00:50:44,060 --> 00:50:50,860
increases, then the position of the farmers who depend on rainfall, as most farmers in

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00:50:50,860 --> 00:50:57,340
the world do, will become really very bad indeed. And it will become bad because they

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00:50:57,340 --> 00:51:01,740
live from hand to mouth. They live pretty close to a subsistence existence. They have

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00:51:01,740 --> 00:51:05,900
no reserves. So it doesn't matter how many good years they have. If they have one bad

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00:51:05,900 --> 00:51:12,900

year, they starve to death. So really bad years coming even far apart might be of catastrophic

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00:51:19,140 --> 00:51:25,100

significance to farmers in the poor countries. It's also probably true that if areas that

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00:51:25,100 --> 00:51:31,660

become drier, which are already more or less dry, if they become drier, the only hope for

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00:51:31,660 --> 00:51:38,660

people in these countries will be immigration. We're seeing already great movements of immigration

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00:51:38,940 --> 00:51:45,260

in different parts of the world. Mexicans and Latin Americans coming into the United

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00:51:45,260 --> 00:51:52,260

States, probably more than one million a year. Vietnamese and Cambodians and Ethiopians and

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00:51:53,260 --> 00:52:00,020

people from Chad and from upper Volta. People from many parts of the less developed world

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00:52:00,020 --> 00:52:07,020

being driven by hunger or by persecution far from their own countries. These migrations

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00:52:07,340 --> 00:52:14,340

might be, which are so serious now, might be greatly worsened if the climate changes

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00:52:14,820 --> 00:52:21,820

for the worse in terms of rainfall over large parts of the areas of the less developed countries.

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00:52:24,300 --> 00:52:31,300

A case in point being China. China is just barely able to feed itself now. They have

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00:52:31,340 --> 00:52:37,980

so little arable land and so many people that they're on the edge of about half the time

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00:52:37,980 --> 00:52:44,980

of having to import considerable amounts of food grains. If the Chinese climate worsens,

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00:52:46,140 --> 00:52:53,140

if it becomes drier, as is very likely to happen, then the food situation for that quarter

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00:52:54,180 --> 00:53:01,180

of mankind might be greatly worsened. We have, of course, even greater uncertainties when

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00:53:01,980 --> 00:53:08,980

we think about the socialist society's reactions and the political and economic repercussions

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00:53:11,740 --> 00:53:16,940

of such drastic climatic change. I don't want to say anything about those because I don't

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00:53:16,940 --> 00:53:23,940

understand them very well, nor do I know of anybody else who has a clear-cut picture.

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00:53:24,940 --> 00:53:31,940

But what can be done is to think about how one can find out about social reactions and

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00:53:34,020 --> 00:53:41,020

about political reactions and about what can be done on a worldwide basis. Because one

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00:53:42,460 --> 00:53:49,460

of the truly remarkable characteristics of this CO2 problem is that it is planetary.

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00:53:49,940 --> 00:53:55,020

All nations and all peoples on Earth will be affected by the gross climatic changes

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00:53:55,020 --> 00:54:01,020

that are likely to result from a large increase in the CO2 content of the atmosphere. But

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00:54:01,020 --> 00:54:07,020

as I've already pointed out, the effects will not be the same. Some regions and some nations

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00:54:07,020 --> 00:54:11,740

will gain more benefits than harm from an atmospheric CO2 buildup, while the reverse

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00:54:11,740 --> 00:54:17,940

will be true of others. These geographical and national differences will greatly exacerbate

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00:54:17,940 --> 00:54:22,980

the difficulties of arriving at an international consensus on such preventive measures as a

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00:54:22,980 --> 00:54:29,980

sharp worldwide reduction in the consumption of fossil fuels. For example, it's very likely

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00:54:30,540 --> 00:54:37,540

that while we might be seriously damaged in the United States by a drying up of the central

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00:54:38,540 --> 00:54:43,940

and eastern parts of our country, the Soviet Union might greatly benefit because the growing

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00:54:43,940 --> 00:54:48,540

season in Siberia and the northern parts of the Soviet Union might be extended by a

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00:54:48,540 --> 00:54:55,540

month. They're always handicapped by a short growing season by the danger of frost in the

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00:54:55,860 --> 00:55:02,860

fall and the very long time lag in the spring before the crops can germinate. So a 25% longer

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00:55:03,860 --> 00:55:10,860

growing season might enable them to double their food production. Even within one country,

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00:55:16,060 --> 00:55:23,060

parts of India might benefit, other parts might be seriously damaged. And this difference

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00:55:23,860 --> 00:55:29,700

in the effects in different parts of the world will make any kind of worldwide action extremely

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00:55:29,700 --> 00:55:35,900

difficult, particularly a worldwide action to reduce the use of fossil fuels. If we look

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00:55:35,900 --> 00:55:41,980

at the projections of energy use during the next 50 years, the best ones have been made

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00:55:41,980 --> 00:55:48,620

by the International Institute of Applied Systems Analysis, in my opinion, in Geneva,

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00:55:48,620 --> 00:55:55,620

I mean in Vienna. Their estimate is that worldwide use of energy would increase by about a factor

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00:55:56,180 --> 00:56:03,180

of five during the next 50 years. And most of this would almost inevitably come from

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00:56:04,420 --> 00:56:11,420

fossil fuels, particularly for the less developed countries. And this means that the less developed

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00:56:14,220 --> 00:56:21,220

countries are liable to greatly increase their production of carbon dioxide, placing greater

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00:56:25,620 --> 00:56:31,580

emphasis on the short-run benefits of industrialization and increased use of energy than the long-range

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00:56:31,580 --> 00:56:38,580

hazards of the carbon dioxide content. On the other hand, there are only really three

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00:56:40,620 --> 00:56:47,180

countries that can really control what happens in the future. Those are the United States,

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00:56:47,180 --> 00:56:54,180

the Soviet Union, and China. And the reason for that is that about 75% of the entire coal

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00:56:55,060 --> 00:57:02,060

reserves of the world and about 60% or more of the entire fossil fuel reserves of the

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00:57:02,620 --> 00:57:09,620

world exist in those three countries, those three countries to which nature and God have

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00:57:11,620 --> 00:57:18,620

been so kind. And it's really our decision and the Chinese decision and the Soviet decision

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00:57:19,620 --> 00:57:26,620

about what will happen as far as fossil fuel, as far as the major part of the fossil fuel

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00:57:26,700 --> 00:57:33,700

use in the future is concerned. Under the best of circumstances, social and national

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00:57:35,140 --> 00:57:40,500

decisions of this kind could be extremely difficult for political leaders because of

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00:57:40,500 --> 00:57:47,500

the long time horizon and the large uncertainties of the carbon dioxide problem. The dilemma

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00:57:47,820 --> 00:57:53,620

is compounded, however, and this I can't emphasize too strongly, by the fact that decisive

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00:57:53,620 --> 00:57:59,140

action must be taken within the next two or three decades if profound climatic changes

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00:57:59,140 --> 00:58:05,700

50 to 70 years in the future are to be avoided. And the reason for this is obvious when you

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00:58:05,700 --> 00:58:12,700

think about it. It takes about 50 years for a new energy source to penetrate the worldwide

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00:58:12,940 --> 00:58:18,380

market. If we're going to make a transition, for example, from fossil fuels to nuclear

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00:58:18,380 --> 00:58:24,500

energy or to solar energy or to wind energy, if you think about that as a major source

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00:58:24,500 --> 00:58:31,500

of energy during the next 50 years, you better start right now. And certainly we need a 30

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00:58:32,260 --> 00:58:39,260

year lead time for any major change in the sources and uses of energy.

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00:58:40,260 --> 00:58:47,260

Let me conclude by pointing out that the carbon dioxide problem has begun to invade public

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00:58:47,580 --> 00:58:52,900

consciousness at a critical time when the worldwide industrial civilization is beginning

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00:58:52,900 --> 00:58:59,900

to be shaken to its foundations by the disappearance of inexpensive sources of energy. This is

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00:59:01,460 --> 00:59:07,780

a time when critical choices must be made about future sources and uses of energy in

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00:59:07,780 --> 00:59:14,100

realization that all potential energy sources, quite apart from the carbon dioxide problem,

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00:59:14,100 --> 00:59:20,660

have serious social, economic, and environmental liabilities. Government and industry must

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00:59:20,660 --> 00:59:26,420

decide whether to invest vast sums of the order of hundreds and perhaps even thousands

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00:59:26,420 --> 00:59:33,420

of billions of dollars in production of synthetic liquid fuels from coal or oil shale. An equally

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00:59:34,220 --> 00:59:39,540

expensive and widely unpopular alternative is construction of many new nuclear fission

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00:59:39,540 --> 00:59:46,540

plants for generation of electricity or production of secondary fuels. Nuclear fission, as you

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00:59:48,060 --> 00:59:53,540

all know better than I, as a long-term alternative to fossil fuels, depends on development and

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00:59:53,540 --> 00:59:59,180

wide use of nuclear breeder reactors with the concomitant problems of proliferation

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00:59:59,180 --> 01:00:06,180

of atomic weapons materials. Energy conservation is another, at least partial, alternative.

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01:00:06,220 --> 01:00:12,540

Energy now used in transportation can be conserved by large investments in mass transit, with

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01:00:12,540 --> 01:00:18,700

all that these investments imply in the chained structure of cities.
Fundamental decisions

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01:00:18,700 --> 01:00:22,980

must also be made whether to continue present patterns of central station generation of

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01:00:22,980 --> 01:00:29,980

electricity and electrical transmission grids or to develop local community energy sources.

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01:00:30,580 --> 01:00:35,980

In the bitter competition for dwindling energy resources, the less developed countries without

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01:00:35,980 --> 01:00:42,220

fossil fuel reserves of their own will inevitably be the losers. This means, as I pointed out,

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01:00:42,220 --> 01:00:48,260

most of mankind. Even development assistance from the rich countries to the poor ones is

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01:00:48,260 --> 01:00:54,100

jeopardized because development of the poor countries will inevitably lead both to great

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01:00:54,100 --> 01:00:59,820

increases in their demand for energy and, perhaps even worse, in their ability to compete

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01:00:59,820 --> 01:01:06,820

with the present industrialized countries to acquire fuels in international markets.

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01:01:10,340 --> 01:01:17,340

Nothing very cheering about this report on the carbon dioxide problem, but I guess I

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01:01:18,380 --> 01:01:24,620
am convinced that, for the moment at least, it's not a very cheerful
world. I think I've

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01:01:24,620 --> 01:01:28,860
talked long enough, but I would be very glad to answer questions if that
is your custom

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01:01:28,860 --> 01:01:35,860
here. I'm better at answering questions than I am at talking.

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01:01:35,860 --> 01:01:42,860
You can go back to 1957 and use the techniques for projecting the effects
of an increase

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01:01:42,860 --> 01:01:49,860
in fossil fuels. What would they say about 1980 when it reached 6%?

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01:01:49,860 --> 01:01:56,340
Well, they'd say if we used those techniques in 1957, it would have come
out pretty much

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01:01:56,340 --> 01:02:03,340
as it did come out, because by 1957, we were experiencing about a 4%
increase in use of

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01:02:03,340 --> 01:02:03,860
fossil fuels.

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01:02:03,860 --> 01:02:08,860
Would they predict any kind of change by now?

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01:02:09,860 --> 01:02:15,860
No, because the natural excursions of climate are of the order of three
or four tenths of

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01:02:15,860 --> 01:02:22,860
a degree centigrade. I started to talk about Mr. Callender and his
explanation of the 70-year-long

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01:02:25,860 --> 01:02:32,860
rise in temperature from 1870 to 1940 as being due to carbon dioxide.
That was a very nice

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01:02:32,860 --> 01:02:39,860

hypothesis, except ever since 1940, the temperature has been decreasing.
So the climatologists

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01:02:39,860 --> 01:02:46,860

are hard put to understand just what has happened.

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01:02:46,860 --> 01:02:53,860

What rate is carbon dioxide going from the upper oceans to the deep
oceans?

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01:02:53,860 --> 01:02:57,860

How fast does the carbon get into the deep ocean? Is that your question?

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01:02:58,860 --> 01:03:05,860

The best evidence for the rate at which CO₂ gets into the ocean is
something called the

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01:03:05,860 --> 01:03:12,860

Seuss effect, which is the decrease in carbon-14 in tree rings from 1850
to 1950, before the

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01:03:14,860 --> 01:03:21,860

first big atomic weapons tests. Those show about a 2% or a little bit
more decrease in

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01:03:22,860 --> 01:03:29,860

carbon-14 content. With any reasonable size of the biosphere, you can't
get very much

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01:03:31,860 --> 01:03:38,860

of the total fossil fuel CO₂ into the ocean. We have a pretty good idea
of the size of

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01:03:41,860 --> 01:03:48,860

the biosphere. The only possibility for more getting into the ocean is
something that I

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01:03:49,860 --> 01:03:56,860

certainly shouldn't mention, although I don't believe it very likely, and
that is that phytoplankton,

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01:03:59,860 --> 01:04:06,860

organisms living in the upper water layers, might die and their bodies descend to the

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01:04:06,860 --> 01:04:12,860

intermediate depths by gravity. This would be sort of a gravitational pump, pumping carbon

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01:04:12,860 --> 01:04:17,860

into the intermediate depths, not by the circulation of the water, but by a completely

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01:04:17,860 --> 01:04:23,860

different mechanism, where the organic matter is oxidized and carbon dioxide is produced.

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01:04:23,860 --> 01:04:29,860

We know that the intermediate depths contain so much carbon dioxide that if you could just

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01:04:29,860 --> 01:04:35,860

stir the ocean up and overturn it, the atmospheric CO₂ would about double. That may very well

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01:04:35,860 --> 01:04:42,860

have happened at the end of the last ice age, at the time of the climatic optimum.

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01:04:43,860 --> 01:04:48,860

The other reason why we don't think so much CO₂ is getting into the deep sea is that the

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01:04:48,860 --> 01:04:53,860

ocean is stratified, and the communication between the deep waters and the waters near

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01:04:53,860 --> 01:05:00,860

the surface is generally believed to be quite slow.

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01:05:01,860 --> 01:05:08,860

Don't you have a lot of history on the carbon dioxide content from ice cores from deep down

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01:05:08,860 --> 01:05:10,860
in the Antarctic and the Arctic?

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01:05:10,860 --> 01:05:16,860
Yes, that's true, and what that seems to indicate is that the atmospheric carbon dioxide content

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01:05:16,860 --> 01:05:23,860
was less during the ice age. It's not quite clear what, at least I haven't seen any good

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01:05:24,860 --> 01:05:30,860
information on what happened at the end of the ice age. Maybe I'm not old Quranic.

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01:05:30,860 --> 01:05:38,860
Over the last few hundred years, are there pretty good year-by-year samples of the air?

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01:05:41,860 --> 01:05:46,860
I don't think those are very good, and the reason is that there are so many factors involved

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01:05:46,860 --> 01:05:53,860
in capturing carbon dioxide in ice. There have been big uncertainties that have been

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01:05:53,860 --> 01:05:55,860
raised about that.

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01:05:58,860 --> 01:06:03,860
How much of the CO2 comes from the clearing of land for agricultural purposes and the

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01:06:03,860 --> 01:06:05,860
oxidation in the soil?

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01:06:05,860 --> 01:06:12,860
What I said was that probably about a third of the total has come that way since 1850,

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01:06:12,860 --> 01:06:19,860

and the reason I say that is that if we look at the total amount of farmland on the earth,

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01:06:19,860 --> 01:06:25,860

I'm convinced that it's increased by about a factor of three since 1850, and a large

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01:06:25,860 --> 01:06:30,860

percentage of that additional farmland must have come from the clearing of forests. The

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01:06:30,860 --> 01:06:35,860

reason I think that farmland has increased by a factor of three is that the population

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01:06:35,860 --> 01:06:41,860

has increased by a factor of three since 1850, and until after World War II, there was hardly

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01:06:41,860 --> 01:06:47,860

any increase in yields, so it must have been just spreading out on more land. And since

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01:06:47,860 --> 01:06:53,860

1950, it's still true that in most of the poor countries, where population growth has

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01:06:53,860 --> 01:07:00,860

been very rapid, that they've accommodated themselves, or they've been able to increase

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01:07:00,860 --> 01:07:08,860

their food production to accommodate the growing population by cutting down forests, spreading

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01:07:08,860 --> 01:07:13,860

out on more land. The principal reason why the forests in low latitudes are diminishing

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01:07:13,860 --> 01:07:20,860

is because of the desperate search of poor people for farmland. There are other reasons,

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01:07:20,860 --> 01:07:22,860
but that's the principal one.

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01:07:22,860 --> 01:07:26,860
And the farm crops don't compensate in terms of use of CO2?

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01:07:26,860 --> 01:07:32,860
No, because you see, you store wood, you store carbon in wood. You don't store carbon in

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01:07:33,860 --> 01:07:41,860
crops, since you eat them and you put the carbon dioxide right back in the air. You

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01:07:41,860 --> 01:07:48,860
have a big storehouse of carbon in the wood of forest trees, something like 750 gigatons,

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01:07:48,860 --> 01:07:55,860
considerably more than there is in the atmosphere. But essentially no storage in agricultural

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01:07:55,860 --> 01:07:56,860
lands.

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01:07:57,860 --> 01:08:07,860
Would the prospect of intensive reforestation reduce the CO2 problem?

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01:08:07,860 --> 01:08:12,860
Sure. That's been advocated by Freeman Dyson, in fact. In fact, he wants to go one step

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01:08:12,860 --> 01:08:21,860
further. He wants not only to grow a lot more forests, trees, but also to cut down the trees

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01:08:21,860 --> 01:08:24,860
and store them someplace, and grow some more.

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01:08:24,860 --> 01:08:30,860
Well, nothing, you know, Freeman Dyson is a very smart guy, and nothing that he says

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01:08:30,860 --> 01:08:38,860

shouldn't be taken seriously. I think myself that this is a second best solution, and if

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01:08:38,860 --> 01:08:44,860

you're going to cut down, if you're going to store carbon in forest trees, the best

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01:08:44,860 --> 01:08:50,860

thing to do is to use the forest trees as a source of energy. Substitute wood for fossil

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01:08:50,860 --> 01:08:51,860

fuels.

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01:08:51,860 --> 01:09:02,860

Does the increase in CO₂ cause a possible shift in the pH in the ocean, or is it sufficiently

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01:09:02,860 --> 01:09:03,860

low?

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01:09:03,860 --> 01:09:10,860

It certainly will reduce the pH of freshwater lakes, and of freshwaters in general. Although

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01:09:10,860 --> 01:09:19,860

the pH of most terrestrial waters seems to be determined by organic processes in the

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01:09:19,860 --> 01:09:30,860

soil and in the water, which tend to lower the pH below that which you'd expect from

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01:09:30,860 --> 01:09:35,860

equilibrium with atmospheric carbon dioxide. In the ocean, you might get a doubling or

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01:09:35,860 --> 01:09:39,860

quadrupling of atmospheric carbon dioxide, but it would result in a significant decrease

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01:09:39,860 --> 01:09:47,860
in pH. Well, I guess it just would, period, because of the change in equilibrium.

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01:09:48,860 --> 01:09:52,860
I'm sorry to hear that.

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01:09:52,860 --> 01:09:57,860
Over the near range, could that CO2 increase cause a possible shift in the ocean?

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01:09:57,860 --> 01:10:00,860
Did you ask about the range of changes?

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01:10:00,860 --> 01:10:06,860
No. Could the change in CO2 cause the ecology of the ocean to start to change?

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01:10:06,860 --> 01:10:11,860
Sure. The pH changes have a profound effect on the phytoplankton.

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01:10:12,860 --> 01:10:18,860
Would it increase or decrease the phytoplankton?

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01:10:18,860 --> 01:10:26,860
It would improve the ecology in the sense that some forms would benefit and others would

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01:10:26,860 --> 01:10:34,860
be damaged. As you know, if you look into these matters, there's a sequence of events

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01:10:34,860 --> 01:10:40,860
in ocean plankton during the year. First things that come are the diatoms, and later

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01:10:40,860 --> 01:10:48,860
the dinoflagellates. These are due to changes in the minor constituents of the sea. The

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01:10:48,860 --> 01:10:56,860
same kind of differences in succession or succession problems would occur with a change

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01:10:56,860 --> 01:11:00,860

in pH. But I wouldn't be able to predict what they would be.

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01:11:04,860 --> 01:11:10,860

What research is the government doing to see if a large-scale sin fuel project is really

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01:11:10,860 --> 01:11:13,860

a good idea?

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01:11:13,860 --> 01:11:19,860

Your question is, what is the government doing to make sin fuels or to not make sin fuels?

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01:11:19,860 --> 01:11:23,860

No, no. To see if it's really a good idea. What's the best in the CO2 problem?

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01:11:23,860 --> 01:11:27,860

Well, one can say that from the point of view of the CO2 problem, it's a bad idea.

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01:11:27,860 --> 01:11:35,860

And the reason is that the CO2 per BTU coming out of synthetic fuels will be about three

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01:11:35,860 --> 01:11:37,860

times what it is from natural gas.

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01:11:37,860 --> 01:11:43,860

But is the government embarking on a project, on a study, to see if a cut in the factors

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01:11:43,860 --> 01:11:46,860

is going to cause a justice problem?

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01:11:46,860 --> 01:11:52,860

Well, we're not doing it quite like that. But the DOE has a major program on the study

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01:11:52,860 --> 01:12:00,860

of all aspects of the carbon dioxide problem, including its social and economic effects.

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01:12:00,860 --> 01:12:05,860

So they're looking at it in that indirect way, knowing full well that if you go to a

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01:12:05,860 --> 01:12:11,860

full-scale synthetic fuel economy, it will rapidly increase the production of CO₂.

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01:12:12,860 --> 01:12:21,860

Did I hear you say that although this carbon dioxide has increased, that the temperature

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01:12:21,860 --> 01:12:23,860

has dropped?

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01:12:23,860 --> 01:12:27,860

Since 1940, it's dropped about two or three-tenths of a degree centigrade.

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01:12:27,860 --> 01:12:36,860

And the explanation that I think is reasonable is that the climate is noisy, and that if

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01:12:36,860 --> 01:12:41,860

you just kept the carbon dioxide content constant, didn't change it at all, you'd

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01:12:41,860 --> 01:12:47,860

get, we know over the past several hundred years, you'd get changes of the order of half

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01:12:47,860 --> 01:12:53,860

a degree to three-quarters of a degree, with a time constant of decades.

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01:12:53,860 --> 01:12:58,860

For example, during the 18th century, the beginning of the 19th century, it was very

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01:12:58,860 --> 01:13:08,860

cold. It was very cold in the 17th century. And it was very warm, let's say, at the time

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01:13:08,860 --> 01:13:14,860

that Greenland was settled and that Iceland was settled. And along about the 13th century,

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01:13:14,860 --> 01:13:18,860

it became quite cold again. It was called the Little Ice Age.

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01:13:18,860 --> 01:13:24,860

Hans Brinker and his silver skates, for example, are an example of this other, this more recent

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01:13:25,860 --> 01:13:28,860

cold period in Europe.

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01:13:30,860 --> 01:13:34,860

How long did we have to wait before you could predict a change in the climate?

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01:13:39,860 --> 01:13:46,860

Well, what the conventional wisdom is that we ought to know pretty well within the next

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01:13:46,860 --> 01:13:54,860

20 years, if the fossil fuel consumption continues to increase exponentially.

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01:13:56,860 --> 01:14:02,860

Because we would be using about twice as much fossil fuels in the year 2000 as we are now

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01:14:02,860 --> 01:14:05,860

if that exponential increase continues.

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01:14:06,860 --> 01:14:10,860

Well, I think we, perhaps, with that question, we ought to thank...

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01:14:10,860 --> 01:14:11,860

Thank you.

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01:14:40,860 --> 01:14:41,860
Thank you.